

How to Set Up Quick and Easy File Sharing With Samba

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What you will need

You don't need much to get Samba up and running:

- A Windows workgroup.
- A Linux machine.
- A bit of time.

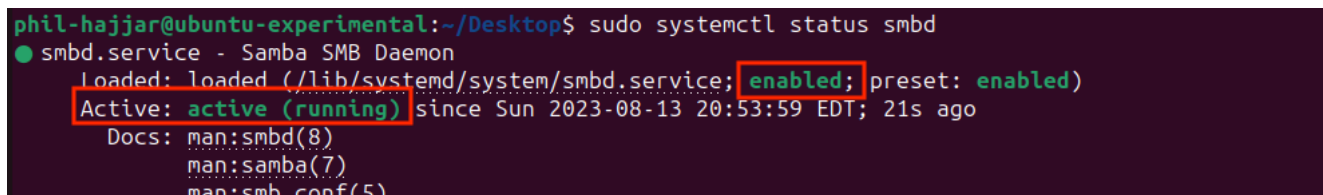
I'll demonstrate this on Ubuntu Desktop 23.04, but the process works the same on most Linux distributions. First, we'll create a folder that will allow anonymous sharing across your workgroup and then create a password-protected share.

Note: You should alter the instructions according to your distribution of choice.

How to install Samba

1. On your Linux machine, open a terminal window.
2. Install the necessary software with the command: `sudo apt-get install -y samba`
3. Type your sudo password, and hit Enter.
4. Allow the installation to complete.
5. Use the command `sudo systemctl status smbd` to check if the Samba service is started and enabled, so it will start on boot (**Figure A**).

Figure A



```
phil-hajjar@ubuntu-experimental:~/Desktop$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; preset: enabled)
   Active: active (running) since Sun 2023-08-13 20:53:59 EDT; 21s ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
```

The command line will indicate whether Samba has been started and enabled.

6. If Samba is not started, use the command `sudo systemctl start smbd` to start it.
7. If Samba is not enabled, use the command `sudo systemctl enable smbd` to enable it.

That's it — Samba will install and start.

How to configure Samba

The main configuration file for Samba is `/etc/samba/smb.conf`. Many will advise you to back up that file and create a new file with specific contents. However, I suggest using this file, as it is better tuned for the release of Samba you've installed.

1. Back up the default configuration file

To that end, make a copy of the default configuration file, so you can safely edit the original and always have a working copy to fall back on. To back up the configuration file:

1. Issue the command: `sudo cp -pf /etc/samba/smb.conf /etc/samba/smb.conf.bak`
2. Now, open the `/etc/samba/smb.conf` file in your favorite text editor, and prepare to make some changes.
3. Look for this line: `workgroup = WORKGROUP`
4. Change `WORKGROUP` to reflect your network needs.

The next section you need to edit is way down in the Share Definitions.

2. Create a directory in Linux you want to share

Before you configure anything with Samba, you will need to create a directory on your Linux server that you will want to share. For this example, we will use `/srv/samba`. To create this directory:

1. Use the command: `sudo mkdir /srv/samba`
2. Now, make that directory world writable and world browsable via: `sudo chmod a+rwX /srv/samba`
3. Go to the bottom of the file, and add the following:

[Anonymous]

```
path = /srv/samba
browseable = yes
writable = yes
read only = no
```

```
guest ok = yes
```

4. Save that file, and restart Samba with this command: `sudo systemctl restart smbd`

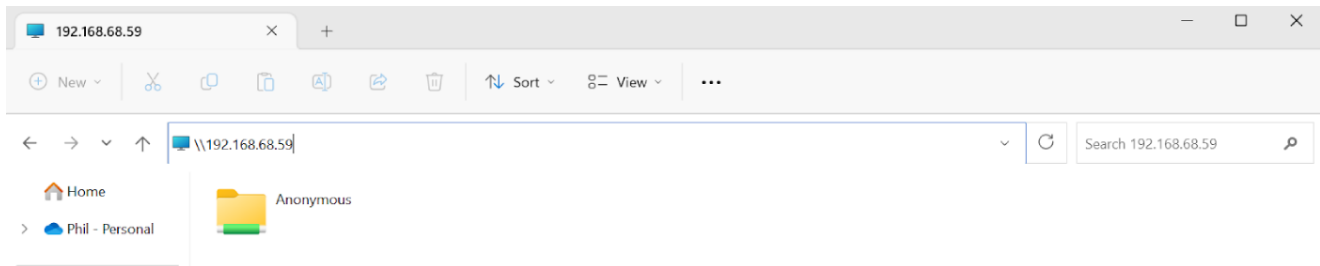
You should be able to reach those shares from any machine on your network.

3. Test the share access

Since we set that share as anonymous, users won't have to log in to access the files and folders within. To test this, you will need the IP address of your Linux server. You can get this via the `ip a` command:

1. On any Windows machine on your network, browse to the shared file using File Explorer, with the syntax \\your-server-ip in the address bar and hitting the Enter key, as shown below (**Figure B**).

Figure B



Adding password-secured shares

We've just added an anonymous share that anyone could access. If you want to add a folder (we'll use /samba/shares as an example) that is password protected, follow these steps:

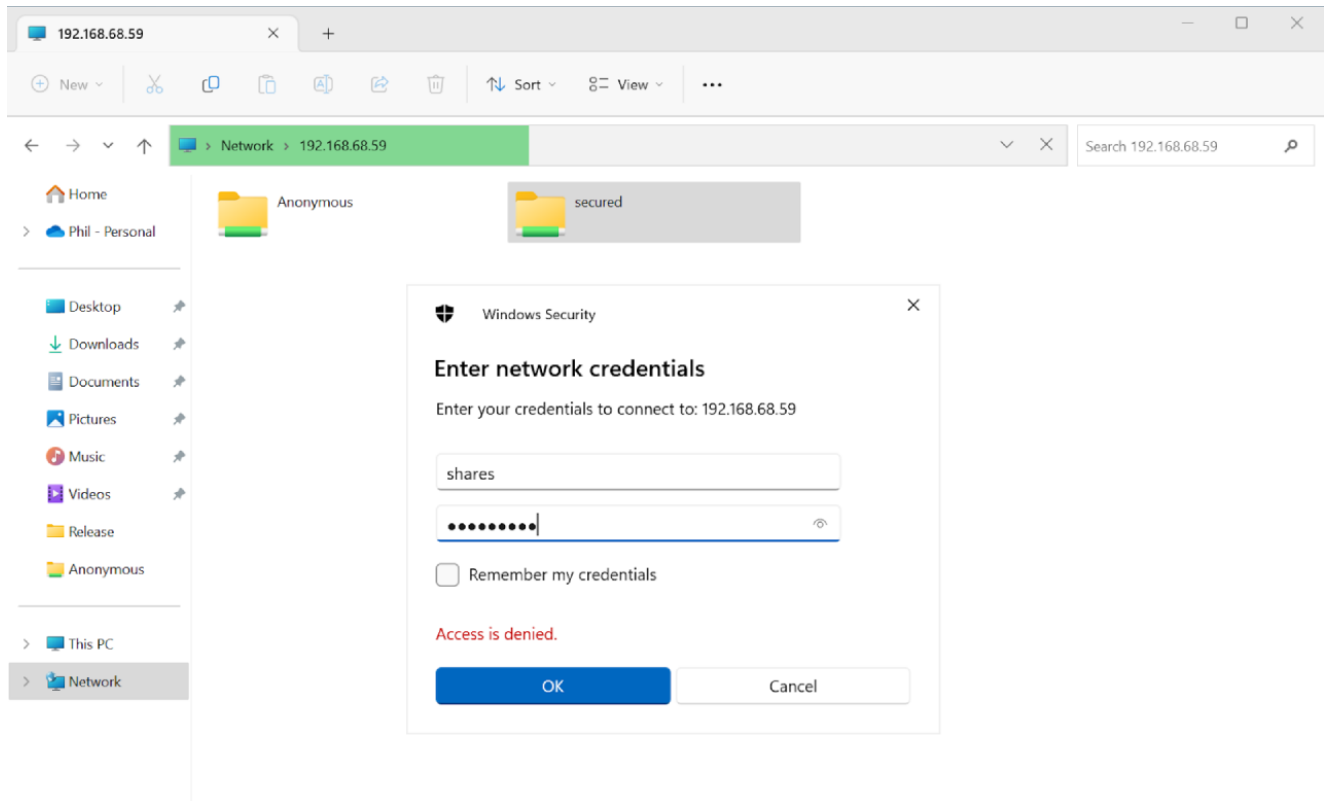
1. Open a terminal window on your Samba server.
2. Create a new group with the command `sudo addgroup smbgrp`.
3. Create a new user with the command `sudo useradd shares -G smbgrp`.
4. Create a Samba password for the user with the command `sudo smbpasswd -a shares`.
5. Type and verify a password for the user.
6. Create the folder with the command `sudo mkdir -p /srv/samba-secured`.
7. Change the permissions of the folder with the command `sudo chmod -R 0770 /srv/samba-secured`.
8. Change the ownership of the folder with the command `sudo chown root:smbgrp /srv/samba-secured`.
9. Now, open the /etc/samba/smb.conf file, and add the following under the Share Definitions or at the bottom of the file:

[SECURED]

```
path = /srv/samba-secured
valid users = @smbgrp
browseable = yes
writable = yes
read only = no
```

10. Save the file, and restart Samba with the command `sudo service smb restart`.

You now have a password-protected Samba share ready to use. Anyone that needs access to the share will log in with username shares and the password you set when you issued the command `sudo smbpasswd -a shares`.



One important note is that if you connected to the anonymous share above, Windows may not let you log in to the secured share. To remedy that, reboot the Windows computer, and try again.